Newton's Laws: Measured with a Sensor Cart

U3 / S3.1

NAME _____ DATE _____ PERIOD _____

GROUP MEMBERS _____

GOALS

- Measure the acceleration of a coasting cart on a level track and report it as a mean with an uncertainty.
- Measure the small force it takes to tow the cart at constant velocity, and connect it to friction.
- Graph force against acceleration and read the cart's mass from the slope, then check it against the balance.
- Measure both forces in one interaction at the same instant, with two force sensors hooked directly together, and compare them.

PURPOSE

You are going to test all three of Newton's laws with numbers instead of taking them on faith. Day 1 is the first and second laws. Day 2 is the third law, with two force sensors hooked directly together.

BACKGROUND

Net force, not force. Newton's laws are about the **net force** $\sum F$, the sum of every force on the object with signs included. A moving object does not need a force to keep moving, only to *change* its velocity. **In Parts A and B, positive is the direction you push or pull the cart. In Part C, positive is toward the person holding the separate sensor.**

> NEWTON'S FIRST LAW
> if $\sum F = 0$, then $a = 0$ · velocity stays constant, including at zero

Mass is the slope. The cart records the pull on its force sensor and its own acceleration at the same moment. On a graph of force (y) against acceleration (x), the points fall on a line whose slope is the mass.

> NEWTON'S SECOND LAW · AND THE SLOPE OF A FORCE–ACCELERATION GRAPH
> $\sum F = ma$ · $N = kg \cdot m/s^2$ slope $= \Delta F \div \Delta a =$ mass · units: $N \div (m/s^2) = kg$

Forces come in pairs. When A pulls on B, B pulls on A with a force of the same size in the opposite direction. The two forces act on *different* objects, so they never cancel. On Day 2 they show up as mirror-image curves: equal size at every instant, opposite sign.

> NEWTON'S THIRD LAW
> $F_{A \text{ on } B} = -F_{B \text{ on } A}$

> MEAN AND UNCERTAINTY (HALF THE RANGE)
> value = mean ± (largest − smallest) ÷ 2

> PERCENT ERROR
> percent error = (| experimental − accepted | ÷ accepted) × 100

> PERCENT DIFFERENCE — WHEN THERE IS NO ACCEPTED VALUE
> percent difference = (| A − B | ÷ [(A + B) ÷ 2]) × 100

SAFETY

- No chemicals, glassware or heat are used in this lab. Follow the room's standing eye-protection rule.
- A cart that rolls off the end of the track falls to the floor. Put an end stop on **both** ends of the track before the first run, and catch the cart by hand before it reaches the stop.
- Fingers get pinched between a bumper and the front of a force sensor. In Part C, hold each sensor by its body, never by its front face, and keep fingers out of the gap.
- A hanging mass can drop when the string slips or the cart reaches the pulley. If you do the extension, keep hands and feet out from under the mass and stop the cart before the mass reaches the floor.
- Pull smoothly and gently. A hard yank makes the cart outrun your hand and slam into the end stop. Nothing in this lab needs a big force.

PRE-LAB — DUE AT THE START OF DAY 1

Answer all of these in your lab notebook, numbered 1 through 9, before you walk in. Do not write on this handout. You will not be allowed to start the lab without the pre-lab finished, because most of these are the parts people get wrong when they wing it.

1. A cart rolls along a level track at a constant 0.40 m/s. What is the net force ∑F on it? Explain using Newton's first law, and say why "the push from earlier is still acting" is wrong.
2. On a graph of force (y-axis, N) against acceleration (x-axis, m/s²), what physical quantity is the slope? Show how the units N ÷ (m/s²) reduce to kg.
3. Your hand sensor pulls directly on the cart's hook with +2.0 N. What force does the cart exert on your sensor, with sign? If the two forces are equal and opposite, why doesn't the cart just sit still?
4. Read Part A, step 3. What do you do to the force sensor before every run, and what must not be pulling on the hook while you do it?
5. Read Part A. Which region of the velocity–time graph do you fit a line to, and which region do you leave out?
6. Read Part B. What is the accepted value you compare the slope to, and how do you get it once a mass bar is on the cart?
7. Read Part C, step 2. Why do both sensors report a pull with the same sign before you reverse one, and what signs should they show after?
8. Safety: where do the end stops go, and how do you hold the sensors during the Part C bumper push? Say exactly where your fingers go.
9. Set up your 3 data tables in your notebook. Copy them exactly as shown, in ink, leaving enough room to write in each blank. Units go in the row labels. Do not plan to add them later.

TABLE 1 · PART A — FIRST LAW
Cart mass, balance (kg)
Push trials 1–3: slope of v–t, a (m/s²)
Mean a ± uncertainty (m/s²)
Tow trials 1–3: mean force (N)
Tow trials 1–3: mean a (m/s²)
Mean tow force ± uncertainty (N)

TABLE 2 · PART B — SECOND LAW
Cart alone: F–a slope (kg)
Cart alone: F–a intercept (N)
Cart + bar mass, balance (kg)
Cart + bar: F–a slope (kg)
Cart + bar: F–a intercept (N)
Percent error, each run (%)

TABLE 3 · PART C — THIRD LAW
Trials C1–C3: peak F on cart (N)
Trials C1–C3: peak F on hand sensor (N)
Percent difference, each trial (%)
C1 stretch: mean F on cart (N)
C1 stretch: mean a (m/s²)
m·a for the stretch (N)

If you write a number wrong during the lab, draw one line through it and write the correct value beside it. Never scribble it out and never erase. I need to see what you originally recorded.

MATERIALS

Per group: Go Direct Sensor Cart · Dynamics Track with an end stop on each end · device with Graphical Analysis · balance · one cart mass bar · string, about 40 cm · **Day 2 only:** one Go Direct force sensor with its hook and bumper (no string) · **Extension only:** track pulley and hanging masses

PROCEDURE

Part A — First Law: No Net Force, No Acceleration

DAY 1 · ABOUT 20 MIN

1. Set the track on the bench with an end stop on each end. Level it.
   - Set the cart at rest in the middle. If it starts to roll, shim the low end with folded paper until it stays put.
   - Give a gentle push each way and compare how fast it slows. If one direction slows faster, the track tilts. Shim and check again.
2. Mass the cart on the balance and record every digit it shows, in kg. Turn the cart on and connect it to Graphical Analysis. Display velocity, acceleration and force against time. Use the acceleration the cart calculates from its wheel, not the accelerometer, in every part. The wheel measures motion along the track; the accelerometer also picks up any tilt.
3. With nothing pulling on the hook, zero the force sensor. Do this before every run, in every part.
4. Push trials, three times:
   - Start data collection. Give the cart one short push and let go. Let it coast for about 2 s, then catch it before the end stop.
   - On the v–t graph, select only the stretch after your hand left the cart. Leave out the push and the catch. Apply a linear fit and record the slope as a, with its sign.
5. Tow trials, three times:
   - Tie the string to the force sensor hook. Zero the sensor with the string hanging slack.
   - Start data collection and tow the cart by the string, level with the track, at a slow, steady pace for about 1 m.
   - Select the stretch where velocity is flat. Record the mean force and the mean acceleration for that stretch.

Part B — Second Law: Mass from a Graph

DAY 1 · ABOUT 20 MIN

1. Keep the string on the hook. Positive is the direction you pull. Zero the force sensor.
2. Start data collection. Pull the cart along the track for about 1–1.5 s, starting gentle and getting firmer, then ease off before the end stop. Keep the string level. Catch the cart.
3. Make a graph of force against acceleration, with acceleration on the x-axis. Select only the stretch where the cart is moving in the + direction, and leave out the catch. Apply a linear fit. Record the slope (kg) and the intercept (N).
   - If the slope comes out negative, the force and acceleration signs disagree. Check the sign convention with me before you take more data.
4. Add one mass bar. Mass the cart and bar together on the balance and record it. Repeat steps 1–3. The slope should go up by about the mass you added.
5. *Extension, if I give you a pulley:* replace your hand with a string over the pulley to a small hanging mass. The force sensor still reads the string tension and the cart still reads a, so each run adds a point to your F–a graph.

Part C — Third Law: Two Sensors, One Interaction

DAY 2 · ABOUT 20 MIN

1. Connect the cart and the separate force sensor to Graphical Analysis so they record together. With nothing pulling on either hook, zero both sensors.

2. Set the sign convention. Positive is toward the person holding the separate sensor.
   - Hook the separate sensor's hook directly onto the cart's hook. No string: a string puts a third object between the two sensors.
   - Give one gentle test tug. Both sensors will read a pull with the **same** sign. That is expected: each one measures the force on its own hook along its own axis, and the two face each other.
   - Reverse the sign of the separate sensor (I will show you how), or plan to multiply its force column by −1. Now both readings are in track coordinates: a pull reads **positive** on the cart and **negative** on the separate sensor.
3. Trial C1, steady pull. Start data collection and pull the cart toward you by the hand sensor, speeding it up gently to a slow speed over about 0.5 m, then ease off so the catch is gentle.
4. Trial C2, tugs. Start data collection and give the cart four or five gentle tugs, easing off between them.
5. For each trial, show both force readings on the same graph. Record the peak force on each sensor, with sign. Sketch the two curves in your notebook with labeled axes, and check whether they mirror each other at every instant, not just at the peak.
6. Trial C3, bumper push. Swap the hook on the separate sensor for its bumper. Zero both sensors. Keep the sign reversal. Press the bumper against the cart's force sensor and push the cart a short way along the track. Catch it before the end stop. Record both peaks.
7. Go back to Trial C1. Select a stretch where the cart is speeding up steadily. Record the mean force on the cart and the mean acceleration for that stretch.
8. Use the rest of the period for the post-lab analysis.

DATA TABLES

You built these in your pre-lab. Have them open before you start data collection. Fill them in **as you take each run**, not at the end of the period from memory. Record every value with units and a sign. Graphs you sketch go right below the table they belong to.

DISPOSAL AND CLEANUP — LAST FIVE MINUTES, BOTH DAYS

1. Untie the string, unhook the sensors, take the mass bar off, and return them to the tray. Nothing goes in the trash.
2. Turn off the cart and the separate force sensor and return them where you got them.
3. Leave the track flat on the bench with both end stops on. Pulley and hanging masses go back in their box.
4. **Wash all glassware with the Alconox solution in the blue spray bottle.** Spray inside, scrub with a brush, and rinse thoroughly — soap left in a beaker changes the next measurement made in it.
5. Rinse with distilled water and invert everything on the rack to dry.
6. Wipe your bench, push in your stool, and wash your hands. **Goggles come off last**, once the whole room is finished — not once you are.

POST-LAB ANALYSIS

Answer these in your lab notebook, numbered 1 through 9. Show every calculation with units and signs. An answer with no work and no units earns partial credit at best.

1. Report the coasting cart's acceleration as mean ± uncertainty. Is it consistent with zero? It will not be exactly zero. Name the force responsible and say which direction it points. PART A

2. Multiply the cart's mass by the size of its mean coasting acceleration. Compare it with your mean tow force. Use Newton's first and second laws to explain why the two should be close. If your tow reading was lost in the sensor's noise, say so, and say which value is the better friction. PART A

3. Calculate percent error for each F–a slope. The accepted value is the balance mass for that run. PART B

4. How much did the slope go up when you added the bar, and how does that compare with the bar's mass? Your fit line probably did not pass through the origin. Explain what the intercept represents, and compare it with your friction value from Part A. PART B · PART A

5. Calculate percent difference between the two peak forces for each trial. Use the size of each force and leave the signs out of this calculation. Then explain what the opposite signs and the mirror-image curves tell you, and why these two forces do not cancel. PART C

6. For the C1 stretch, calculate m·a for the cart. Compare it with $\sum F = F - f$, where F is the mean force on the cart and f is your friction value from Part A. Explain why you have to subtract f. PART C · PART A

7. A group forgot to zero the force sensor in Part B, and it read +0.15 N with nothing on the hook. Does that change the slope of their F–a graph, the intercept, or both? Explain what happens to their mass and their percent error. ERROR ANALYSIS

8. Suppose the track was not level and the cart coasted slightly downhill in Part A. What would that do to the sign and size of the push-trial slope? Name the step that was meant to catch it. ERROR ANALYSIS

9. In Section 1.3 you learned that the slope of a v–t graph is acceleration. Point to the place in this lab where you used that, and say what quantity you would get from the *area* under the same graph. U1 / S1.3